

Holonomic 2D Hovercraft MPC (MPPI)

Planar double integrator with l_1 thrust cap: sampling-based MPPI steers to a draggable goal while soft-penalizing circular obstacles — rollout fan and best predicted path drawn live.

around fixed **circular rocks** (exponential “skirt” when outside plus a strong **quadratic penalty** if the robot disk penetrates the inflated obstacle boundary). The weighted-average control update matches MPPI’s information-theoretic reweighting; the simulator applies the **first mean action** until the next replan at interval Δt . Purple fan trajectories (optional when **K** is not huge) visualize the ensemble; the **green** polyline is the **lowest-cost** sample path from the last plan. Layout presets create a **narrow corridor** or an **offset pillar field** so students can trade **horizon**, **temperature λ** , **noise σ** , and **obstacle weights** against conservatism and compute budget.

Who it's for: Students who finished the pendulum MPPI demo and want the same sampling intuition on a **2D navigation** task before touching full nonholonomic models or QP-based MPC.

KEY TERMS

- Holonomic robot
- Double integrator
- Model Predictive Path Integral (MPPI)
- Trajectory sampling
- Soft collision constraints
- Obstacle avoidance
- Warm-started control sequence

How it works

Holonomic 2D “hovercraft” with $\ddot{\mathbf{x}} = \mathbf{u}_x, \ddot{\mathbf{y}} = \mathbf{u}_y$ and bounded **L2** thrust: **MPPI** rolls **K** noisy acceleration trajectories over horizon **H**, adds **obstacle skirts** around circular rocks, and reweights the mean control each step — same sampling spirit as **engineering/mpc-pendulum**, now with **goal seeking** and **collision avoidance** in the plane.

Key equations

$\ddot{\mathbf{x}} = \mathbf{u}_x, \ddot{\mathbf{y}} = \mathbf{u}_y$, with $\|(\mathbf{u}_x, \mathbf{u}_y)\|_2 \leq u_{\text{max}}$ – holonomic double integrator (hovercraft abstraction).
MPPI: sample noisy accelerations along the horizon, score rollouts with goal, speed, effort, and obstacle soft penalties, then exponential weights update the mean control sequence.

Frequently asked questions

Why soft obstacles instead of hard “if collision discard rollout”?

Hard rejection creates **discontinuous** costs and brittle gradients for sampling methods—many rollouts die in useless configurations. Smooth penalties keep **grad-like** information in the exponential weights while still exploding cost on deep penetration.

Is this literally a hovercraft dynamics model?

Only as a **cartoon**: real hovercraft have attitude–thrust coupling, fan dynamics, and ground effect. The **double integrator with $\|\mathbf{u}\|_2$ cap** is the standard **holonomic** surrogate used in curricula to isolate **MPC in the plane**.

Why does the robot sometimes scrape obstacles despite MPC?

Soft costs allow **trade-offs**: if **q_pos** dominates or **horizon** is short, the planner may briefly accept small penetration to cut a corner. Raise **q_crash**, lengthen **H**, or increase **R** on control to be more cautious.

Does disabling the rollout fan change the controller?

No—it only hides the **purple** ensemble for performance. The **green** best-path preview and applied **u** are unchanged.

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Sine $\rightarrow$ samples $\rightarrow$ quantization $\rightarrow$ ZOH; Nyquist and optional aliasing demo.